DATA ITEM DESCRIPTION

Title: SAFETY ASSESSMENT REPORT (SAR)

Number: DI-SAFT-80102D

Approved Date: 20250715

AMSC Number: F10580

Limitation: N/A

DTIC Applicable: No

GIDEP Applicable: No

Preparing Activity: 40 (AFMC/SE)

Project Number: SAFT-2025-002

Applicable Forms: N/A

Use/Relationship: The Safety Assessment Report (SAR) is a comprehensive evaluation of the safety risks being assumed prior to test, operation of the system or at contract completion. The SAR summarizes the safety features across the design. It also summarizes the hazards and associated hazard controls present in the system being acquired.

a. This Data Item Description (DID) contains the content and format preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

b. This DID is related to DI-SAFT-80101, *System Safety Hazard Analysis Report (SSHAR)*; DI-SAFT-80105, *System Safety Program Progress Report (SSPPR)*; and DI-SAFT-80106, *Health Hazard Assessment Report (HHAR),* DI-SAFT-82085, *Hazard Tracking System (HTS) Data*.

(Copies of these DIDs are available online at https://quicksearch.dla.mil.)

c. This DID supersedes DI-SAFT-80102C.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The SAR shall be in the contractor's format.

3. Content. The SAR shall include the following information:

3.1 Introduction. State, in narrative form, the purpose of the SAR.

3.2 System description. This section shall be developed by referencing other program documentation such as technical manuals, system safety program plan, system specification, etc., and shall include the following:

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3.2.1. The purpose and intended use of the system.

3.2.2. A brief historical summary of system development.

3.2.3. A brief description of the system and its components. Include name, type, model number, and general physical characteristics of the overall system and its major subsystems and components.

3.2.4. A brief overview of the software architecture and its roles, emphasizing safety critical elements within the system.

3.2.4.1. A brief overview of how the program defines units of software and means used to identify where in the software a discussion relates to.

3.2.4.2. A brief description of the software environment and which higher order language it is written in.

3.2.5. As applicable, a description of any other system(s) which will be tested or operated in combination with this system. This would include System of Systems (SoS) functionality.

3.2.6. As applicable, either photos, charts, flow/functional diagrams, sketches, or schematics to support the system description, test, or operation.

3.3 System operations. This section shall include:

3.3.1. A description or reference of the procedures for operating, testing and maintaining the system. Discuss the safety design features and controls incorporated into the system for the various modes of operation as they relate to the operating procedures.

3.3.2. A description of any special safety procedures needed to assure safe operations, test and maintenance, including emergency procedures.

3.3.3. A description of anticipated operating environments, and any specific skills required for safe operation, test, maintenance, transportation or disposal.

3.3.4. A description of any special facility/infrastructure requirements, support equipment or personal equipment to support the system.

3.4 Systems safety engineering. This section shall include:

3.4.1. A summary or reference of the safety criteria and methodology used to classify and rank hazardous conditions.

3.4.2. A description of or reference to the analyses and tests performed to identify hazardous conditions inherent in the system.

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3.4.2.1. A list of all hazards (including hardware and software) by subsystem or major component level that have been identified and considered from the inception of the program in an appendix to this SAR.

3.4.2.1.1. A discussion of the hazards and the actions that have been taken to eliminate or control these items.

3.4.2.1.2. A discussion of the effects of these controls on the probability of occurrence and severity level of the potential mishaps.

3.4.2.1.3. A discussion of the residual risks that remain after the controls are applied or for which no controls could be applied

3.4.2.2. A discussion of or reference to the results of tests conducted to validate safety criteria requirements and analyses.

3.4.3. A description of reference to how software is partitioned into the units /function/task of software as documented within the SAR.

3.4.3.1. For each unit of software, list the Software Control Category (SCC) per MIL-STD-882E(C1) Table IV with the accompanying supporting rationale.

3.4.3.2. For each unit of software, list the worst credible Severity per MIL-STD-882E(C1) Table I with the accompanying supporting rationale.

3.4.3.3. For each unit of software, list the derived Software Criticality Index (SwCI) per MIL-STD-882E(C1) Table V and the accompanying tier of Level of Rigor (LOR).

3.4.3.4. For each tier of LOR, summarize all the LOR task activities.

3.4.3.5. Identify when MIL-STD-882E(C1) Table VI will be queried. Typical program timelines may include prior to Preliminary Design Review (PDR), prior to Critical Design Review (CDR), prior to formal testing, prior to fielding, and periodically during sustainment.

3.4.3.6. For each unit of software, summarize applicable LOR task activities.

3.4.3.7. For each unit of software for each LOR task activity, provide or cite the reference to all program artifacts required to show compliance with the LOR task activity.

3.4.3.8. For each unit of software for each LOR task activity, provide the determination from MIL-STD-882E(C1) Table VI if a risk exists or not, and what are the recommended actions for the government program management to take.

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3.5 Conclusions and recommendations. This section shall include:

3.5.1. A short assessment of the results of the safety program efforts.

3.5.1.1. A list of all significant hazards along with specific safety recommendations or precautions required to ensure the safety of personnel and property. The list of hazards will be categorized as to whether or not they may be expected under normal or abnormal operating conditions.

3.5.1.2. A summary status of all LOR task activities for all units of software. This status shall highlight which LOR task activities are on track to be completed on time, which LOR task activities have been completed, which LOR task activities are not on track to be completed on time, and which LOR task activities are determined to be non-compliant.

3.5.2. For all hazardous materials generated by or used in the system:

3.5.2.1. Materiel identification as to type, quantity, and potential hazards.

3.5.2.2. Safety precautions and procedures necessary during use, storage, transportation, and disposal.

3.5.2.3. A copy of the Safety Data Sheet (SDS) as required.

3.5.3. A summary of all explosive hazard classifications generated under the program's contract.

3.5.4. A statement that the system does not contain or generate hazardous materials (i.e. explosive, toxic, radioactive, carcinogenic, etc.).

3.5.5. A statement signed by the contractor system safety manager and the program manager stating that all identified hazards have been eliminated or controlled, and that the system is ready to test, operate, or proceed to the next acquisition phase. In addition, include recommendations applicable to the safe hazards at the interface of this system with the other system(s).

3.6 References. A list of all pertinent references, such as test reports, preliminary operating manuals and maintenance manuals.

End of DI-SAFT-80102D

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